EVAN ZHANG

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EDUCATION

The Harker School, San Jose, CA **Expected Graduation: May 2026**

- **ACT:** 36
- **Advanced Topics Courses:** AP Calc BC, AP Physics 1, AP Chem, AP Euro, AP CSA, AP Chinese, AP Physics C Mech, AP Physics C E&M, AP US History, AP Stats, AP Env. Science, AP US Gov, AP English Lit, Honors Linear Algebra, Honors Multivariable Calculus, Honors Discrete Math

RESEARCH & ENTREPRENEURSHIP

***Researcher & Start-up Founder & CEO*: Enviro-AI, developing "SPIRo", G10-12** **June 2023 - Present**

- Pneumatic and modular soft robot, SPIRo, with an ensemble learning algorithm for inspecting gas leaks.
- Founded VC-backed start-up Enviro-AI, secured the 1517 Fund Medici Grant, Nonprovisional Patent (publication expected in Jan 2026). 2 first-author publications, 2 second-author publications

***Research Intern*, Chortos Lab at Purdue University, G11-Present** **June 2024 - Present**

- Developed AI-controlled HASEL actuators for cardiac assist devices, integrating pressure sensing and FEM modeling to optimize control and response. Pioneered a Direct Ink Write (DIW) 3D printing method utilizing thermally elongated nozzles.
- ***Start-up Founder & CEO*: Pulsar Robotics**, developing soft robotic ventricular assist devices (VADs)

***Paid Research Summer Intern*, The Faboratory, Yale University, G11-12 Summer** **July - August 2025**

- Built a shape-morphing amphibious robot with variable stiffness limbs, enabling adaptive locomotion.
- Proposed a deep-water capable robotic turtle, designed high-pressure waterproof casings for motors.

***Mechanical Engineering Intern*, Schnitzer Lab at Stanford University, G9-Present** **Dec 2022 - Present**

- Designed circuit schematics and PCBs for sensing and control of experimental apparatus used across the lab.
- Fabricated and built a 1P microscope w/ 3-degree-of-freedom adjustable side illumination.

PUBLICATIONS

- Omnimobile and Load Capable Pneumatic Soft Robot for Gas Distribution Pipeline Inspection (IEEE Robosoft 2025, April 2025, Full Oral Talk) Evan Zhang, Eddie Zhang
- Gas Pipeline Leakage Detection Based on Multiple Multimodal Deep Feature Selections and Optimized Deep Forest Classifier (Frontiers in Environmental Science, April 2025) Eddie Zhang, Evan Zhang
- Mitigating Low-Speed Crawling and Jitter in Telescopic Hydraulic Cylinders Through Stick-Slip Dynamics Analysis and Friction Reduction Strategies (Journal of Tribology, 2025) Zeyu Ma, … Evan Zhang
- SPIRo: An AI-Based Origami-Inspired Soft Robot with Multidimensional Locomotion and Multimodal Data Analysis for Infrastructure Assessments (IEEE CCDC, May 2024) Evan Zhang, Eddie Zhang
- Development of A Multimodal Deep Feature Fusion with Ensemble Learning Architecture for Real-Time Gas Leak Detection (IEEE ICMI, April 2024, Won Best Paper Award) Eddie Zhang, Evan Zhang
- 3D-Printed HASEL Actuators With Integrated Fluidic Piezoresistive Pressure Sensor for Cardiac Assist Applications (ICMSR 2025, December 2025, In-Press) Evan Zhang, Alex Chortos
- Towards a Deep Water Amphibious Robotic Turtle (IEEE Robosoft 2026, April 2026, In-Press) Luis Ramirez, … Evan Zhang, … Rebecca Kramer-Bottiglio
- Patent #19/274609: SPIRo: An AI-Based Origami-Inspired Pneumatic Soft Robot with Multidimensional Locomotion and Multimodal Deep Feature Selection and Fusion with an Improved Deep Forest Classifier Architecture for Real-Time Gas Pipeline Leak Detection

OTHER LEADERSHIP EXPERIENCES

***Co-founder*, Cream of the Crop 501 (c) 3 Non-Profit Organization, G9-Present** **Aug 2021 - Present**

- Delivered classes and summer workshops in software, robotics, 3D design, and RC aircraft for 40+ students.
- Donated $4.5K in STEM kits, led workshops at Title I schools in the Redwood City District, providing 60+ students with access to applied STEM learning. Hosted a district-wide LEGO Robotics League and classes.
- Tennis clinics and lessons for 20+ beginner and intermediate players.

***Captain and Coach*, Harker Varsity Debate Team, G9-Present** **Aug 2022 - Present**
- Ranked Top 80 nationally, qualifying for the Tournament of Champions in Public Forum debate.
- Coached incoming 10+ freshmen and middle school students at practices, intramurals, and tournaments.

***Intern,* Harvard Undergraduate Tech Ventures Summer Program, G11-12 Summer** **June - July 2025**
- Led a team integrating Gemini to generate suggestions for energy companies for a YC-backed startup.
- Learned about topics such as product-market fit, go-to-market strategy, and fundamentals of startup creation.

***Founder and President*, Harker 3D Printing and Design Club, G11-Present** **June 2024 - Present**
- Founded and scaled the club to 60+ members, organized challenges and competitions within the club.
- Taught classes on CAD Design and Print Slicing, trained members on the 3D printer.

***Co-President,* Harker Aerospace Club, G9-Present** **Aug 2022 - Present**
- Founded and led 2 teams to compete in the American Rocketry Challenge, grew the club to 70+ members.
- Secured $2.5K in institutional funding through formal pitches and proposals to school administration.

***Mechanical Director,* FTC 11311/ *Design Team*, Harker FRC 1072, G9-11** **Aug 2021 - August 2024**
- Led a team of 10 in the design and fabrication of robots, trained 5 new members on CAD & CNC
- Contributed to FIRST World Championship qualification, NorCal Regional Finalist placement

***Founder*, Enabling The Future Chapter, G10-Present** **Nov 2024 - Present**
- Established a 4-member team designing prosthetics and published designs for use by volunteers.

***Founder*, 3DSolutions.shop, G10-Present** **July 2024 - Present**
- 3D printing business, designing and fabricating custom hardware, generating $1K in revenue.

HONORS/ACADEMIC ACHIEVEMENTS

- **Coca-Cola Scholars *Semifinalist*** (September 2025)
- **Regeneron Science Talent Search** (STS) ***Top 300 Scholar*** (January 2026)
- **International Science and Engineering Fair** (ISEF) Finalist & ***Grand Prize 4th Place*** (May 2025)
- **Conrad Challenge *Pete Conrad Scholar***, Energy and Environment Category (April 2025)
- **California State Science Fair** 2025 Category 5th Place (April 2025), Kindeva Innovator Award
- **Diamond Challenge** Finalist 2025 (April 2025)
- **Tournament of Champions (TOC)** Qualifier in Public Forum Debate, ranked T80 Nationally (April 2025)
- **Sigma Xi 2025 Student Research Showcase** 1st Place in High School Division (April 2025)
- **Nor. Cal. Junior Science and Humanities Symposium** 2025, Second Place in Chemistry (February 2025)
- **Synopsys Science Fair** Santa Clara County Grand Award, Category 1st award, ASEI special award (2025)
- **1517 Fund, Medici Grant** of $1k for SPIRo & Enviro-AI (2025)
- **National Junior Science and Humanities Symposium** Finalist 2024, National Oral Presentation Project
- **Synopsys Science Fair** Santa Clara County 2024 Category 2nd Award, IEEE special award
- **AIME Qualifier** (2023, 2024) with Distinction (2023), **USACO Silver** Qualifier (2024)
- **TSA TEAMS** Competition Nationals Qualifier (2023, 2025)
- **FRC World Championship Qualifier Team**, FTC NorCal Regionals Finalist (2023)
- **Presidential Volunteer Award** Gold & Bronze (2024, 2023)
- **Scholastic Writing,** Silver Key (2023, 2024), Honorable Mention (2026)

SKILLS

Design (Onshape, Fusion 360, KiCad), Fabrication (3D printing, CNC), Modeling (Abaqus, OpenRocket), Programming (Java, Python, Mathematica, LaTeX, LabVIEW, OpenCV), Control Systems (MyRIO, Arduino)